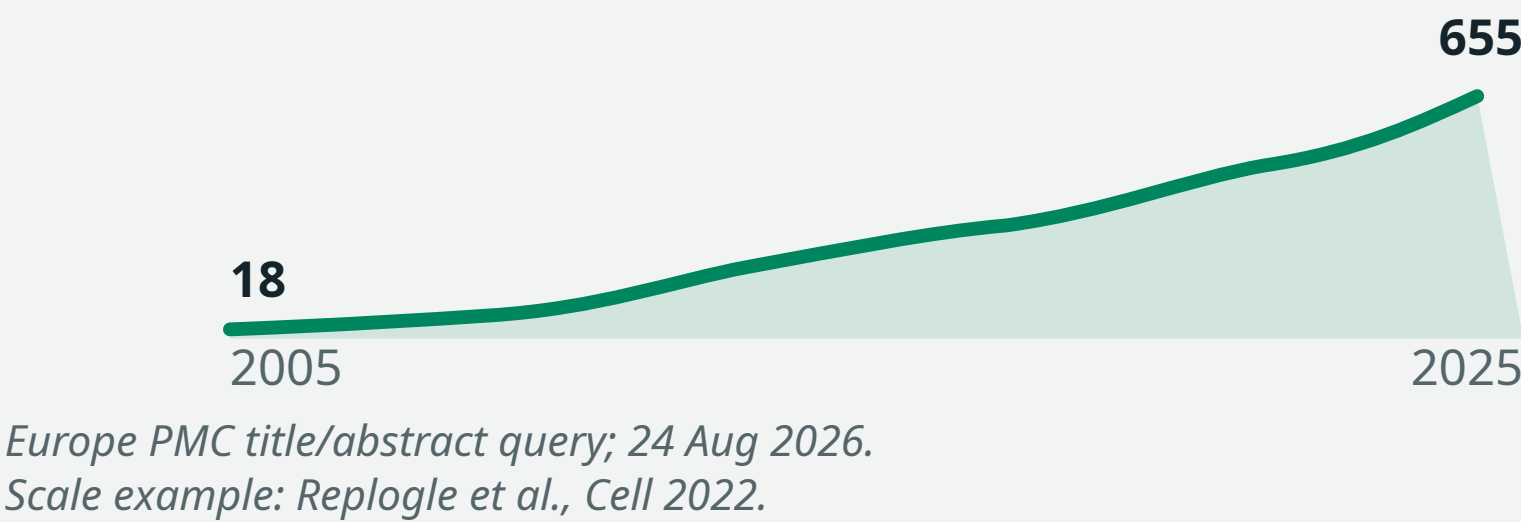


Perturbation evidence is multiplying faster than it can be organised

A genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, while pooled screens test thousands of targets. That scale could reveal robust biology across studies—but identifiers, metadata, assays and processing pipelines differ. Results remain hard to compare or reproduce. The Catalogue supplies a shared schema, stable gene identity and one access path for combining this evidence.

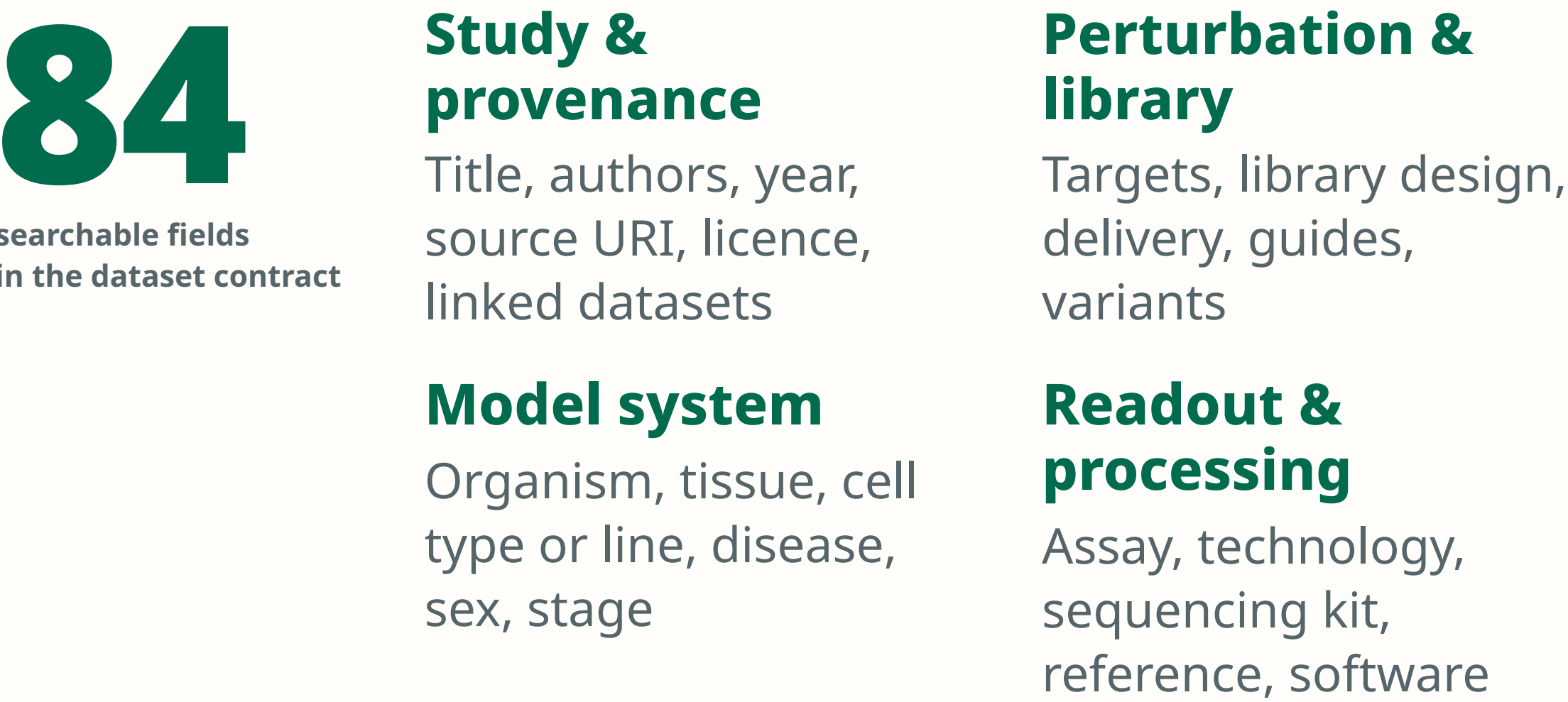
36× growth
in perturbation papers
over two decades



A single catalogue turns scattered evidence into a searchable resource

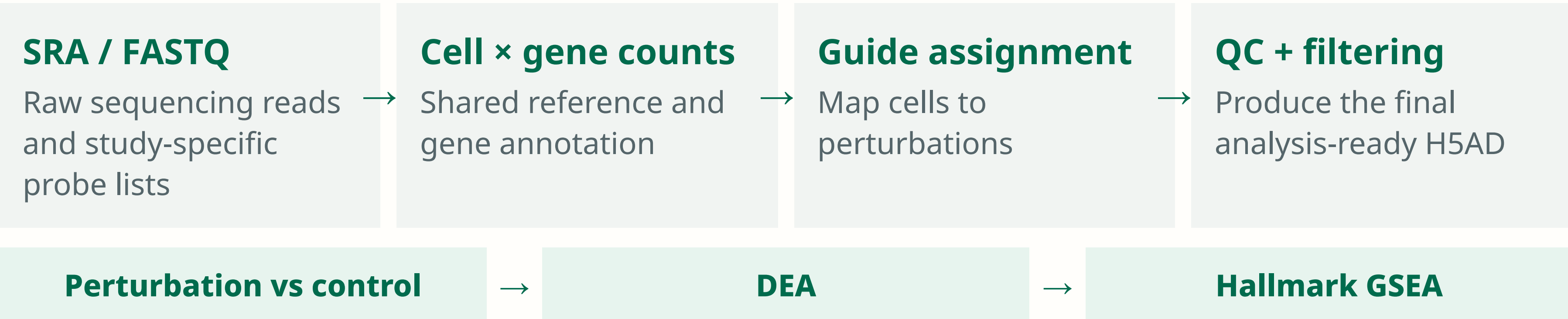


Metadata and reanalysis make results comparable



Perturb-Seq reprocessing starts again from raw reads

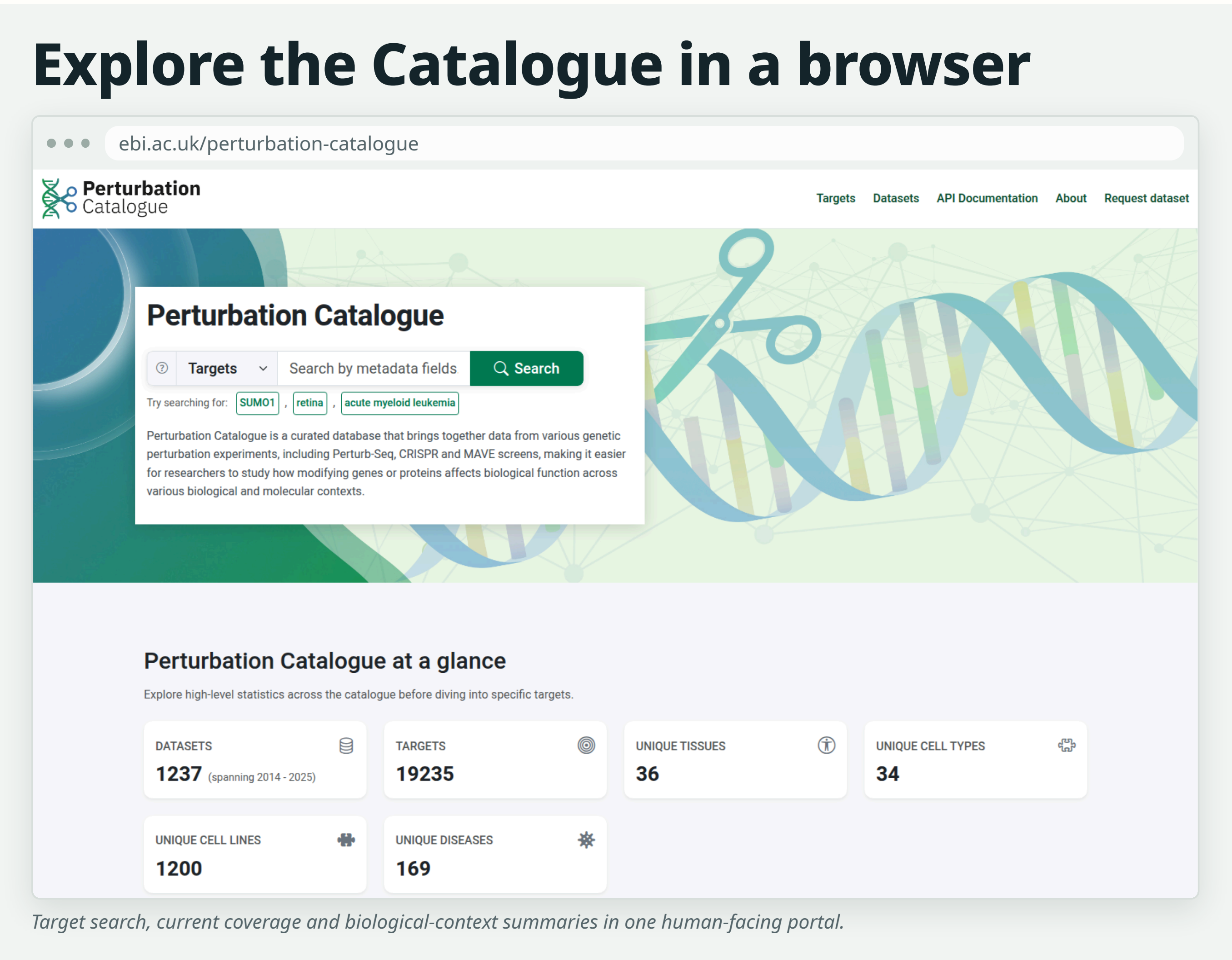
Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.



Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Stable Ensembl gene IDs and ontology-linked entities make the

Explore the Catalogue in a browser



The screenshot shows the Perturbation Catalogue browser interface. It features a search bar with a dropdown menu for 'Targets' and a 'Search' button. Below the search bar, there is a section titled 'Perturbation Catalogue at a glance' which displays high-level statistics across the catalogue. The statistics are: DATASETS (1237, spanning 2014 - 2025), TARGETS (19235), UNIQUE TISSUES (36), UNIQUE CELL TYPES (34), UNIQUE CELL LINES (1200), and UNIQUE DISEASES (169). The interface also includes a 'Try searching for' section with suggestions like 'SUMO1', 'retina', and 'acute myeloid leukemia'.

Target search, current coverage and biological-context summaries in one human-facing portal.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```

```
{
  "total_rows_count": 30078802,
  "results": [{
    "perturbation": {"gene_symbol": "TFAM"},
    "effect": {
      "gene_symbol": "MT-ND5",
      "log2fc": -1.926, "padj": 0.0
    }
  }]
}
```

Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review	Reanalyse more designs and measure quality independently	Publish periodic, versioned releases with explicit provenance
AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.	More kits and experimental structures, author-independent QC and modality-appropriate methods.	Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.

perturbation-catalogue-help@ebi.ac.uk

Explore the Catalogue

